



PRAYAS

2.0 2027

Mathematics

Lecture - 05

Basic Mathematics

By – Pradeep Verma





Topics To Be Covered



- 1 Logarithm Function and Inequalities

① PP-I

9, 10, 11

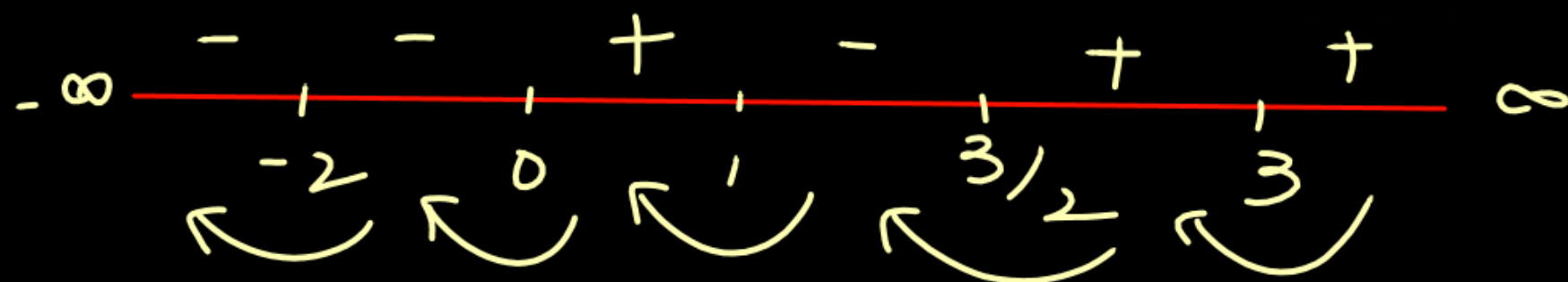
16, 17
13, 14, 18, 20, 21, 22, 24

23



⑨

$$x(x+2)^2(x-1)^5(2x-3)(x-3)^4 \geq 0$$



$$> 0 \Rightarrow x \in (0, 1) \cup \left(\frac{3}{2}, 3\right) \cup (3, \infty)$$

$$\geq 0 \Rightarrow x \in [0, 1] \cup \left[\frac{3}{2}, 3\right] \cup [3, \infty)$$

$$x \in [0, 1] \cup \left[\frac{3}{2}, \infty\right)$$

$$x \in \{-2\} \cup [0, 1] \cup \left[\frac{3}{2}, \infty\right)$$

$$x \in \{-2\} \cup [a, b] \cup [c, \infty)$$

$$a + b + c = 5/2$$

⑩ $\frac{x+1}{x^2+2} > \frac{1}{4}$

$$\frac{x+1}{x^2+2} - \frac{1}{4} > 0$$

$$\frac{4x+4-x^2-2}{4(x^2+2)} > 0$$

$$\begin{aligned} -x^2 + 4x + 2 &> 0 \\ x^2 - 4x - 2 &< 0 \end{aligned}$$

$\alpha < \beta$

$$x^2 - 4x - 2 < 0$$

$$(x - \alpha)'(x - \beta)' < 0$$

$-\infty$ $+$ $-$ $+$ ∞
 α β

$$x \in (\alpha, \beta)$$

$$x^2 - 4x - 2$$

$$x = \frac{4 \pm \sqrt{16 + 8}}{2}$$

$$= \frac{4 \pm 2\sqrt{6}}{2} = 2 + \sqrt{6}, 2 - \sqrt{6}$$

β α

$$x \in (2 - \sqrt{6}, 2 + \sqrt{6})$$

⑪

$$\frac{5 - 4x}{3x^2 - x - 4} - 4 < 0$$

$$\frac{5 - \cancel{4x} - 12x^2 + \cancel{4x} + 16}{3x^2 - x - 4} < 0$$

$$\frac{-12x^2 + 21}{3x^2 - x - 4} < 0$$

$$\frac{-3(4x^2 - 7)}{3x^2 - 4x + 3x - 4} < 0$$

PW

$$\frac{(2x)^2 - (\sqrt{7})^2}{(x+1)(3x-4)} \geq 0$$

$$\frac{(2x - \sqrt{7})(2x + \sqrt{7})}{(x+1)(3x-4)} \geq 0$$

⑬ $\log_{(\sqrt{3}-\sqrt{2})} (5-2\sqrt{6})$

$\rightarrow a^2 + b^2 - 2ab$

$$\begin{aligned} 5 - 2\sqrt{6} &= (\sqrt{2})^2 + (\sqrt{3})^2 - 2\sqrt{2}\sqrt{3} \\ &= (\sqrt{3} - \sqrt{2})^2 \end{aligned}$$

$$= \log_{(\sqrt{3}-\sqrt{2})} (\sqrt{3}-\sqrt{2})^2$$

$$= 2$$

⑭ $\log_3 (\log_2 512)$

$$\log_3 (\log_2 2^9)$$

$$\log_3 3^2$$

$$= 2$$

⑰ $\log_6 18 + \log_6 2 - \log_2 3 - \log_2 3^{-1}$

$$\log_6 6^2 - \cancel{\log_2 3} + \cancel{\log_2 3}$$

$$= 2$$

⑱ $x = \log_{2a} a$

$$y = \log_{3a} 2a, \quad z = \log_{4a} 3a$$

$$xyz - 2yz$$

$$(xyz - 2yz)$$

$$\left(\log_a \log_{2a} \log_{3a} \log_{4a} - 2 \log_{3a} \log_{4a} \right)$$

$$\left(\frac{\log a}{\log 2a} \cdot \frac{\log 2a}{\log 3a} \cdot \frac{\log 3a}{\log 4a} - 2 \frac{\log 2a}{\log 3a} \cdot \frac{\log 3a}{\log 4a} \right)$$

$$\frac{\log a}{\log 4a} - \frac{\log 4a^2}{\log 4a}$$

$$\frac{\log 1/4a}{\log 4a} = \frac{\log (4a)^{-1}}{\log 4a} = -1 \frac{\log 4a}{\log 4a} = -1$$

(21) $N = \sqrt{(2+\sqrt{5}) - \sqrt{(6-3\sqrt{5})} + \sqrt{(14-6\sqrt{5})}}$

$\downarrow$
 $\sqrt{(3^2 + (\sqrt{5})^2 - 2 \cdot 3 \cdot \sqrt{5})}$
 $\downarrow$
 $\sqrt{(3-\sqrt{5})^2}$

$$\begin{aligned}
 N &= \sqrt{(2+\sqrt{5}) - \sqrt{6-3\sqrt{5}} + 3-\sqrt{5}} \\
 &= \sqrt{(2+\sqrt{5}) - \sqrt{9-4\sqrt{5}}} \\
 &= \sqrt{2+\sqrt{5} - \sqrt{(\sqrt{5}-2)^2}}
 \end{aligned}$$



$$\begin{aligned}
 &= \sqrt{2+\sqrt{5} - (\sqrt{5}-2)} \\
 &= \sqrt{4} = 2
 \end{aligned}$$

(23)

$$81^{1/\log_5 3} + 27^{\log_9 36} + 3^{4/\log_9 9} \rightarrow 3^{4 \log_9 9}$$

$$81^{\log_3 5} + 36^{\log_{3^2} 3^3} + (7^4)^{\log_{3^2} 3}$$

$$3^{\log_9 7^4}$$

(24)

$$\frac{\log_a x}{\log_{ab} x} = 4 + K + \log_a b \quad \text{find } K$$

$$\log_a x \cdot \frac{1}{\log_{ab} x} \Rightarrow \frac{1}{\log_a x} \log_a b$$

$$\frac{\log_a b}{\log_x a} = \log_a b$$

$$= \log_a a + \log_a b$$

$$= 1 + \log_a b$$

$$= 4 + \log_a b + K$$

$$4 + K = 1$$

$$K = -3$$

(22)

$${}^4_5 \log_{2^{5/2}} (3 - \sqrt{6}) - 6 \log_{2^3} (\sqrt{3} - \sqrt{2})$$



$${}^4\sqrt{2} = 2^2 \cdot 2^{1/2} = 2^{5/2}$$

$${}^4 \log_{(2^{5/2})^{1/5}} (3 - \sqrt{6}) - \frac{6}{3} \log_2 (\sqrt{3} - \sqrt{2})$$

$${}^4 \log_{2^{1/2}} (3 - \sqrt{6}) - 2 \log_2 (\sqrt{3} - \sqrt{2})$$

$${}^4 \log_2 (3 - \sqrt{6}) - 2 \log_2 (\sqrt{3} - \sqrt{2})$$

$$\log_{a^N} M = \frac{1}{N} \log_a M$$

$$\log_{a^{1/N}} M = \frac{1}{1/N} \log_a M$$

$$= N \log_a M$$

$${}^4_2 (\log_2 (3 - \sqrt{6}) - \log_2 (\sqrt{3} - \sqrt{2}))$$

$${}^4_2 \left(\log_2 \frac{\sqrt{3}(\sqrt{3} - \sqrt{2})}{(\sqrt{3} - \sqrt{2})} \right)$$

$${}^4_2 \log_2 \sqrt{3} = {}^4 \log_2 3 = 3 \log_2 2^2 = 3^2 = 9$$



JEE Wallah
WhatsApp channel



**Scan & Join
the JEE Wallah
WhatsApp Channel
For Important Updates
and Announcements**

Match the column

Column-I		Column-II	
A.	$\log_{\sin 30^\circ}(\cos 60^\circ) + 1$	p.	3
B.	$\log_{4/3}(1.\bar{3}) + 3$	q.	5
C.	$\log_{2-\sqrt{3}}(2 + \sqrt{3}) + 6$	r.	4
D.	$\log_{\tan 20^\circ} \tan 70^\circ + 4$	s.	2
E.	$\log_{40^\circ} \tan 50^\circ$	t.	0
F.	$\log_{0.125}(8) + 8$	u.	-1
G.	$\log_{1.5}(0.\bar{6}) + 9$	v.	8
H.	$\log_{2.25}(0.4)$	w.	7
I.	$\log_{10}(0.\bar{9})$	x.	1

- A** $A \rightarrow q, B \rightarrow p, C \rightarrow s, D \rightarrow v, E \rightarrow u,$
 $F \rightarrow u, G \rightarrow q, H \rightarrow w, I \rightarrow x$
- B** $A \rightarrow s, B \rightarrow r, C \rightarrow q, D \rightarrow p, E \rightarrow x,$
 $F \rightarrow w, G \rightarrow v, H \rightarrow u, I \rightarrow t$
- C** $A \rightarrow s, B \rightarrow v, C \rightarrow t, D \rightarrow p, E \rightarrow t,$
 $F \rightarrow u, G \rightarrow w, H \rightarrow x, I \rightarrow w$
- D** $A \rightarrow q, B \rightarrow s, C \rightarrow r, D \rightarrow v, E \rightarrow u,$
 $F \rightarrow v, G \rightarrow v, H \rightarrow w, I \rightarrow x$

Question

Solve the following logarithmic equations:

$$\log_5(5^{1/x} + 125) = \log_5(6) + 1 + \frac{1}{2x}$$

A $2, 1/2$

B $1, 1/8$

C $\frac{1}{4}, 1/2$

D $1, 2/5$

$$\log_5(5^{1/x} + 125) - \log_5 6 = 1 + \frac{1}{2x}$$

$$\log_5 \left(\frac{5^{1/x} + 125}{6} \right) = 1 + \frac{1}{2x}$$

$$\frac{5^{1/x} + 125}{6} = 5^{1 + 1/2x}$$

$$\frac{5^{1/x} + 125}{6} = 5 \cdot 5^{1/2x}$$

$$5^{1/2x} = t \Rightarrow 5^{1/x} = t^2$$

$$\frac{t^2 + 125}{6} = 5t$$

$$t^2 + 125 = 30t$$

$$t^2 - 30t + 125 = 0$$

$$(t - 25)(t - 5) = 0$$

$$5^{1/2x} = 5^2 \cdot 5^1$$

$$\frac{1}{2x} = 2.1$$

$$x = 1/4, 1/2$$

Question



If $\log_x \log_{18}(\sqrt{2} + \sqrt{8}) = \frac{1}{3}$. Then the value of $1000x$ is equal to

- A** 8
- B** $1/8$
- C** $1/125$
- D** 125

$$\log_{18}(\sqrt{2} + 2\sqrt{2}) = x^{1/3}$$

$$\log_{18}(3\sqrt{2}) = x^{1/3}$$

$$\log_{18}(18^{1/2}) = x^{1/3}$$

$$\frac{1}{2} = x^{1/3} \Rightarrow x = \frac{1}{8}$$

$$1000 \times \frac{1}{8} = 125$$

Question



If $3^{2 \log_3 x} - 2x - 3 = 0$, then the number of values of ' x ' satisfying the equation is

$$3^{\log_3 x^2} - 2x - 3 = 0$$

A Zero

B 1

C 2

D More than 2

$$x^2 - 2x - 3 = 0$$

$$x^2 - 3x + x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = -1, 3$$

The solution set of the system of equation $\log_3 x + \log_3 y = 2 + \log_3 2$ and

$\log_{27}(x + y) = \frac{2}{3}$ is:

☒ **A** {6, 3}

☐ **B** {9, 6}

☐ **C** {6, 12}

☐ **D** {12, 6}

$$\log_3 xy - \log_3 2 = 2$$

$$\log_3 \left(\frac{xy}{2} \right) = 2$$

$$\frac{xy}{2} = 9$$

$$xy = 18$$

$$x + y = (3^{\cancel{3}})^{2/\cancel{3}}$$

$$x + y = 9$$

$$xy = 18$$

Question

The number of real solution/s of the equation

$9^{\log_3(\log_e x)} = \log_e x - (\log_e x)^2 + 1$ is :

- ☐ A 0 $(\log_e x)^{\log_3 3^2} = \dots$
- ☒ B 1 $(\log_e x)^2 = \log_e x - (\log_e x)^2 + 1$
- ☐ C 2 $t^2 = t - t^2 + 1$
- ☐ D 3 $2t^2 - t - 1 = 0$



$$2t^2 - 2t + t - 1 = 0$$

$$2t(t-1) + (t-1) = 0$$

$$(t-1)(2t+1) = 0$$

$$t = 1, -\frac{1}{2}$$

$$\log_e x = 1, -\frac{1}{2}$$

$$x = \boxed{e}, \frac{1}{\sqrt{e}}$$

✓ ✗

Question



The solution set of $\log_{10} 4 + \left(1 + \frac{1}{2x}\right) \log_{10} 3 = \log_{10}((3)^{1/x} + 27)$

A $\left(\frac{1}{2}, \frac{1}{4}\right)$

$$\log_{10} 4 + \log_{10} 3^{1 + \frac{1}{2x}} = \log_{10} (3^{1/x} + 27)$$

B $\left\{\frac{1}{2}, \frac{1}{4}\right\}$

$$\log_{10} (4 \cdot 3^1 \cdot 3^{1/2x}) = \log_{10} (3^{1/x} + 27)$$

C $\left[\frac{1}{2}, \frac{1}{4}\right]$

$$12 \cdot 3^{1/2x} = 3^{1/x} + 27$$

$$3^{1/2x} = t \Rightarrow 3^{1/x} = t^2$$

$$12t = t^2 + 27$$

$$t^2 - 12t + 27 = 0$$

$$t^2 - 9t - 3t + 27 = 0$$

D None of these

$$(t-9)(t-3) = 0$$

$$t = 3, 9$$

$$3^{1/2x} = 3, 9$$

$$\frac{1}{2x} = 1, 2$$

$$\frac{1}{x} = 2, 4$$

$$x = \frac{1}{2}, \frac{1}{4}$$

QUESTIONS [JEE 2011, 3(-1)]



Let (x_0, y_0) be the solution of the following equations

$(2x)^{\ln 2} = (3y)^{\ln 3}$, $3^{\ln x} = 2^{\ln y}$ then x_0 is

A

$\frac{1}{6}$

B

$\frac{1}{3}$

~~C~~

$\frac{1}{2}$

D

6

$$\ln(2x)^{\ln 2} = \ln(3y)^{\ln 3}$$

$$\ln 2 (\ln 2 + \ln x) = \ln 3 (\ln 3 + \ln y)$$

$$\ln^2 2 + \ln 2 \cdot \ln x = \ln^2 3 + \ln 3 \cdot \ln y$$

$$\ln^2 2 - \ln^2 3 = \ln 3 \cdot \ln y - \ln 2 \cdot \ln x$$

$$\ln 3^{\ln x} = \ln 2^{\ln y}$$

$$\ln x \cdot \ln 3 = \ln y \cdot \ln 2$$

$$\ln y = \frac{\ln 3 \cdot \ln x}{\ln 2}$$

$$\ln^2 2 - \ln^2 3 = \ln 3 \left(\frac{\ln 3 \cdot \ln x}{\ln 2} \right) - \ln 2 \cdot \ln x$$

$$(\ln^2 2 - \ln^2 3) = \frac{(\ln^2 3 - \ln^2 2) \cdot \ln x}{\ln 2}$$

$$-1 = \frac{\ln x}{\ln 2}$$

$$\ln x = -\ln 2$$

$$\ln x = \ln 2^{-1}$$

$$x = \frac{1}{2}$$

$$x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}} = \sqrt{2 + x}$$

The value of $6 + \log_{\frac{3}{2}}$

$$\left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \dots}}} \right) \text{ is}$$

A 8

B 1

C 2

D 4

$$x = \frac{1}{3\sqrt{2}} \sqrt{4 - x}$$

$$x^2 = \frac{1}{18} (4 - x)$$

$$18x^2 + x - 4 = 0$$

$$18x^2 + 9x - 8x - 4 = 0$$

$$9x(2x+1) - 4(2x+1) = 0$$

$$(2x+1) \cdot (9x-4) = 0$$

$$x = \left(-\frac{1}{2}\right)^b, \frac{4}{9}$$

$$\begin{aligned} 6 + \log_{\frac{3}{2}} \left(\frac{4}{9}\right) &= 6 + \log_{\left(\frac{3}{2}\right)'} \left(\frac{3}{2}\right)^{-2} \\ &= 6 - 2 \\ &= 4 \end{aligned}$$

QUESTIONS

**If $3^x = 4^{x-1}$, then $x =$

A

$$\frac{2 \log_3 2}{2 \log_3 2 - 1}$$

B

$$\frac{2}{2 - \log_2 3}$$

C

$$\frac{1}{1 - \log_4 3}$$

D

$$\frac{2 \log_2 3}{2 \log_2 3 - 1}$$

$$\log_3 3^x = \log_3 4^{x-1}$$

$$x = (x-1) \log_3 4$$

$$x = (x-1) \log_3 2^2$$

$$x = 2(x-1) \log_3 2$$

$$x = 2x \log_3 2 - 2 \log_3 2$$

$$2 \log_3 2 = x (2 \log_3 2 - 1)$$

$$x = \frac{2 \log_3 2}{2 \log_3 2 - 1}$$

$$x = \frac{2 \log_3 2}{\frac{2 \log_3 2}{\log_2 2} - \frac{1}{\log_2 2}}$$

$$x = \frac{2}{2 - \log_2 3}$$

$$x = \frac{1}{1 - \frac{1}{2} \log_2 3} = \frac{1}{1 - \log_4 3}$$

The value of $((\log_2 9)^2)^{\frac{1}{\log_2(\log_2 9)}} \cdot (\sqrt{7})^{\frac{1}{\log_4 7}}$ is

- A** 2
- B** 4
- C** 8
- D** None



Basic – Inequality



$$\bullet \quad x^2 > a^2 \Rightarrow x > a \cup x < -a \Rightarrow x \in (-\infty, -a) \cup (a, \infty)$$

$$\bullet \quad x^2 < a^2 \Rightarrow -a < x < a \Rightarrow x \in (-a, a)$$

$$\bullet \quad a^2 < x^2 < b^2 \Rightarrow -b < x < -a \cup a < x < b$$
$$x \in (-b, -a) \cup (a, b)$$



Logarithmic Function



➤ **Domain, Range & Graph:**



Log - Inequality



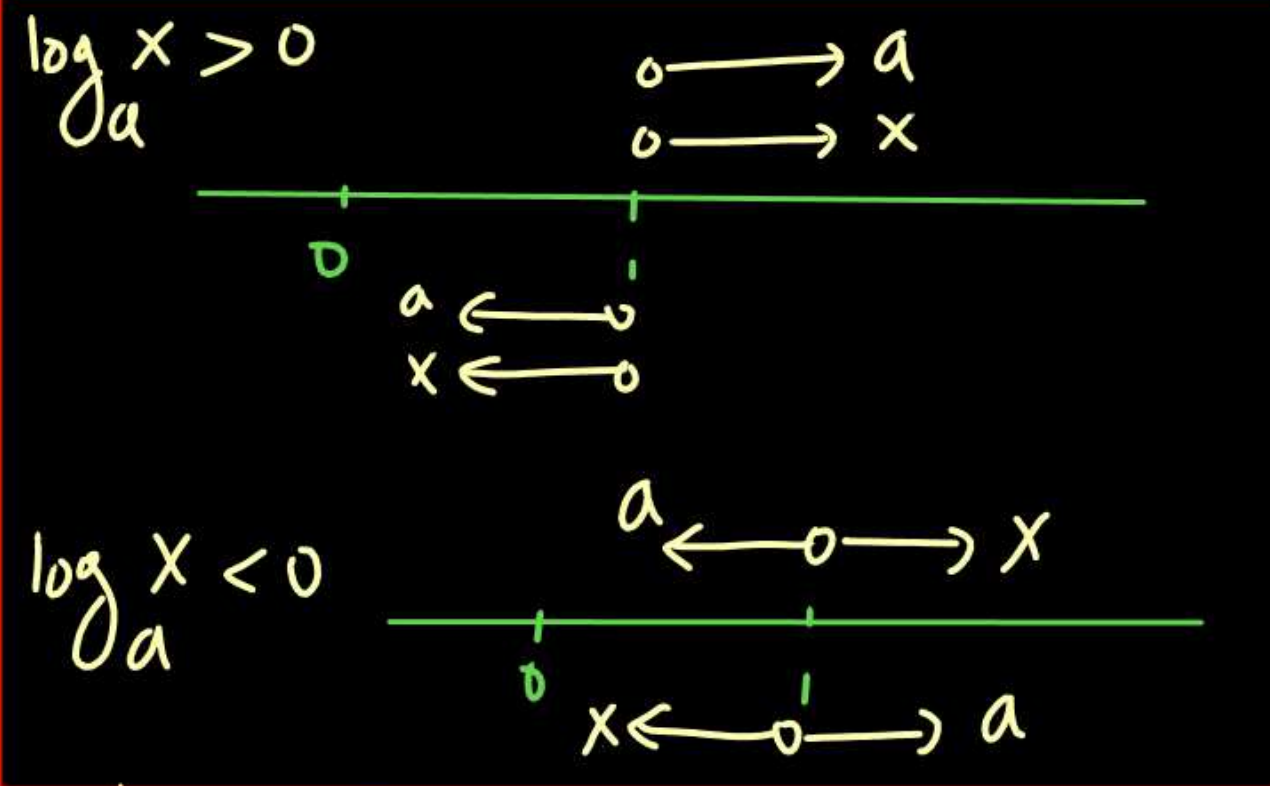
$$\log_a x, x > 0, a > 0, a \neq 1$$

$$x > y > 0 \quad \cdot \quad a > 1$$

$$\star \log_a x > \log_a y$$

$$0 < x < y, \quad 0 < a < 1$$

$$\star \log_a x > k \quad \begin{cases} x > a^k, & a > 1 \\ 0 < x < a^k, & 0 < a < 1 \end{cases}$$



$$\star \log_{1/2} 1/2 = 1, \log_{1/2} 2 = -1, \log_2 2 = 1, \log_2 1/2 = -1$$

Question

$$\log_a x \Rightarrow x > 0, a > 0, a \neq 1$$



Solve following inequality:

$$\log_{0.5} \left(\log_8 \left(\frac{x^2 - 2x}{x - 3} \right) \right) < 0$$

$$\log_{1/2} (\log_8 (t)) < 0$$

$$\textcircled{\text{I}} \quad t > 0$$

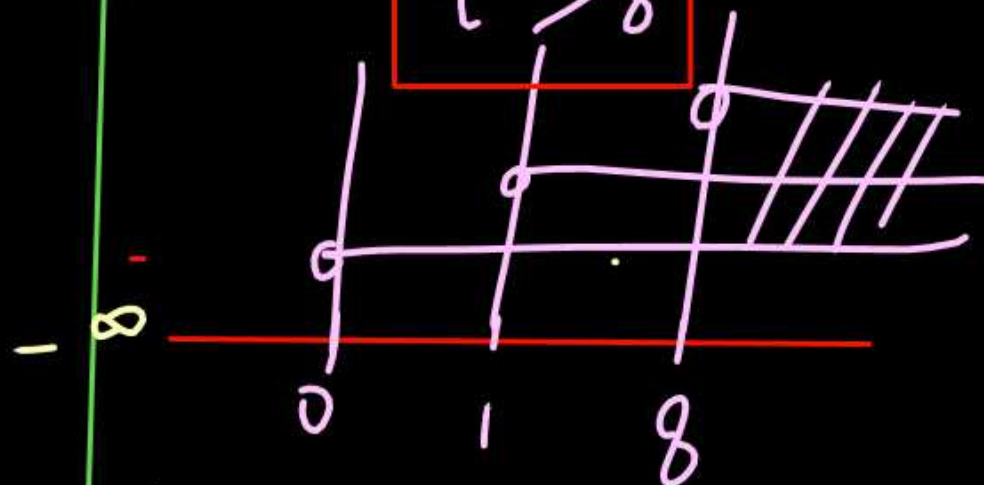
$$\textcircled{\text{II}} \quad \log_8 t > 0 \Rightarrow t > 8^0 \Rightarrow (t > 1)$$

$$\textcircled{\text{III}} \quad \log_{1/2} (\log_8 (t)) < 0$$

$$\log_8 (t) > \left(\frac{1}{2}\right)^0$$

$$\log_8 t > 1$$

$$t > 8$$



$$t > 8$$

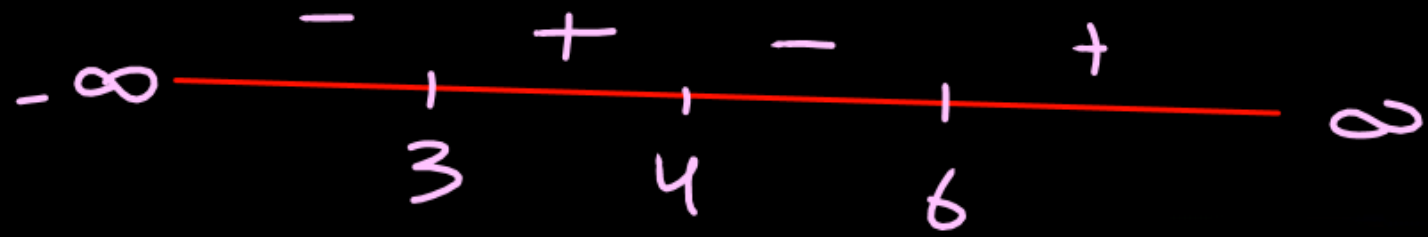
$$\frac{x^2 - 2x}{x - 3} > 8$$

$$\frac{x^2 - 2x}{x - 3} - 8 > 0$$

$$\frac{x^2 - 2x - 8x + 24}{x - 3} > 0$$

$$\frac{x^2 - 6x - 4x + 24}{(x - 3)} > 0$$

$$\frac{(x-4)(x-6)}{(x-3)} > 0$$



$$x \in (3, 4) \cup (6, \infty)$$

Question



Solution set of the inequality $2 - \log_2(x^2 + 3x) \geq 0$ is:

- A** $[-4, 1]$
- ☒ **B** $[-4, -3) \cup (0, 1]$
- C** $(-\infty, -3) \cup (1, \infty)$
- D** $(-\infty, -4) \cup [1, \infty)$

$$2 - \log_2 t \geq 0$$

$$t > 0$$

$$2 \geq \log_2 t$$

$$\log_2 t \leq 2$$

$$t \leq 2^2$$

$$t \leq 4$$

$$0 < t \leq 4$$

$$0 < x^2 + 3x \leq 4$$

$$x^2 + 3x > 0$$

$$x(x+3) > 0$$

$$\begin{array}{c} + \quad - \quad + \\ -3 \quad 0 \quad \infty \end{array}$$

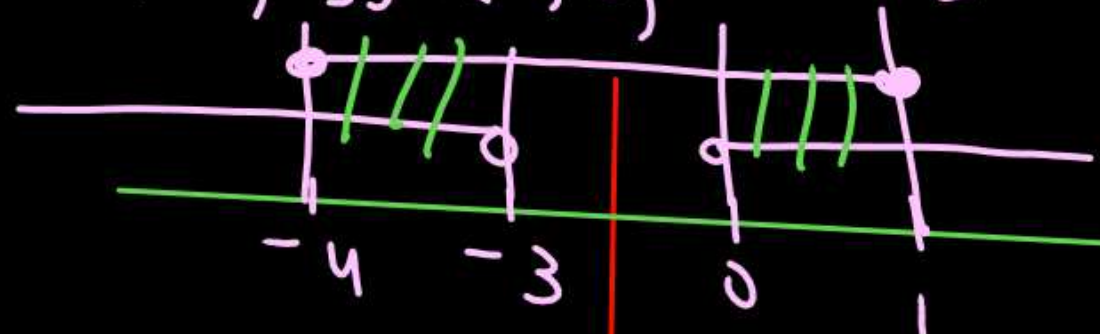
$$x \in (-\infty, -3) \cup (0, \infty)$$

$$x^2 + 3x - 4 \leq 0$$

$$(x+4)(x-1) \leq 0$$

$$\begin{array}{c} + \quad - \quad + \\ -4 \quad 1 \quad \infty \end{array}$$

$$x \in [-4, 1]$$



Question



The set of real values of x satisfying $\log_{1/2}(x^2 - 6x + 12) \geq -2$ is

- A** $(-\infty, 2]$
- B** $[2, 4]$
- C** $[4, +\infty)$
- D** $[3, 8]$

Thank

You